The Impact ofLabor Market Indicators on the Official Exchange Rate in Algeria (1991–2024)

أثر مؤشرات سوق العمل على سعر الصرف الرسمي في الجزائر (1991–2024)

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Abstract:

This study aimed to measure the impact of labor market indicators on the official exchange rate in Algeria 1991-2024. The study relied on a multiple linear regression model.

The results revealed relationships between labor market indicators and the exchange rate. An inverse relationship was observed between both employment and the exchange rate, as well as between general unemployment and the exchange rate. This indicates that improvements in labor market indicators contribute to the stability of the local currency, reflecting the unique impact of these indicators on the Algerian economy.

Keywords: Labor Marke; Exchange Rate; Unemployment; OLS; Algeria.

JELClassificationCodes:E24, F31, C21

ملخص:

قاست الدراسة أثر مؤشرات سوق العمل على سعر الصرف الرسمي في الجزائر خلال الفترة 1991-2024. وقد اعتمدت الدراسة على نموذج الانحدار الخطي المتعدد.

أظهرت النتائج وجود علاقات بين مؤشرات سوق العمل وسعر الصرف، حيث تم تسجيل علاقة عكسية بين كل من التشغيل وسعر الصرف، وكذلك بين البطالة العامة وسعر الصرف، ما يشير إلى أن تحسّن مؤشرات سوق العمل يساهم في استقرار العملة المحلية، مما يعكس خصوصية تأثير هذه المؤشرات على الاقتصاد الجزائري.

كلمات مفتاحية: سوق العمل، سعر الصرف، البطالة، طريقة المربعات الصغرى، الجزائر.

تصنيفات JEL: E24، F31، C21

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1. INTRODUCTION

Employment has emerged as one of the most pressing structural challenges facing Arab economies particularly Algeria in the aftermath of the sharp decline in oil prices during the late 1980s. Despite Algeria's repeated efforts to formulate policies and forward-looking strategies aimed at reducing unemployment and promoting decent work, these attempts have largely fallen short. This shortfall is attributable to a combination of deteriorating macroeconomic conditions, declining investment volumes, and persistent structural imbalances in the economy.

Addressing the employment crisis requires a nuanced and data-driven reassessment of the current labor market dynamics. In this context, the Algerian government should strategically mobilize public spending not only as a short-term stimulus, but as a lever to increase labor force participation and revitalize market activity. Enhancing participation rates, particularly among youth and marginalized groups, has the potential to generate momentum for new investment projects that are labor-intensive by nature. This virtuous cycle can contribute to stabilizing or even positively influencing fluctuations in the official exchange rate, given the interlinkages between employment, production capacity, and external economic balances.

1.1 Problem Statement: As outlined in the introduction regarding the imbalance in the Algerian labor market, the central research problem can be summarized in the following main question:

How have labor market indicators influenced the official exchange rate in Algeria over the period from 1991 to 2024?

From this main question, several sub-questions can be derived:

- What are the indicators and characteristics of the labor market?
- Has the labor market in Algeria reached a level that allows it to positively influence the official exchange rate during the period 1991–2024?

1.2 Research Hypotheses: As preliminary answers to the research problem stated above, the following hypotheses have been adopted:

- In Algeria, employment and unemployment represent two of the most critical metrics used to assess labor market performance.

- Despite progress, Algeria's labor market has not matured enough to play a constructive role in shaping the official exchange rate.

1.3 Research Objectives: This study aimed to present the key concepts related to the labor market. It also sought to measure the impact of labor market indicators on the official exchange rate in Algeria during the period from 1991 to 2024.

1.4 Research Methodology: To answer the research problem and address the components of the study, the inductive approach was followed, using both descriptive and analytical methods for the theoretical part. For the practical part, the quantitative approach was adopted, through the use of a multiple linear regression model. The data used in the study were collected from the World Bank database.

2. Theoretical Approach to Labor Market Indicators

The labor market represents one of the fundamental pillars of any economy, as it reflects the interaction between the supply and demand for labor. Labor market indicators are essential analytical tools for understanding the dynamics of this market, such as unemployment rates, labor force participation rates, employment levels, and wages. To interpret and analyze these indicators scientifically, a set of theoretical approaches has emerged, aiming to understand the economic and social relationships that govern the labor market.

2.1 Definition of the Labor Market: Definitions of the labor market vary widely from one researcher to another and from one institution to another. Among the most significant definitions, we can highlight the following:

- "The labor market is defined as the environment in which workers seek to sell their services to employers, who in turn hire them under mutually agreed-upon terms and conditions. It is also defined as an economic regulatory institution where the supply and demand for labor interact that is, the space in which labor services are bought and sold, and thus priced (belhadri, cherifi, & meddah, 2020, p. 03)."
- "It is the domain that matches workers with jobs, or where labor is exchanged for wages or in-kind compensation. The labor force constitutes the vital resource that feeds the market with workers (djalal & ahmed, 2020, p. 44)."

- "It is a virtual market that represents the supply of human labor capable and willing to work, in contrast to the demands expressed by institutions in various economic and social sectors for vacant or newly created positions. It refers to the supply and demand of human resources (joughrouri, 2023, p. 03)."

As a comprehensive definition, the labor market can be described as a virtual market a type of economic market in which the supply of labor meets the demand for it. It acts as a connecting link between all individuals directly involved in employment.

2.2 Characteristics and Features of the Labor Market: The labor market possesses several distinctive features, among which the following can be highlighted (jedidi, 2016, p. 172):

- **Absence of perfect competition**: This means that there is no single uniform wage in the market for similar types of work;
- **Ease of distinguishing between labor services:** Even when jobs are similar, labor services can be easily differentiated due to factors such as race, age, culture, gender, skin color, religion, etc;
- **Influence of technological advancement:** The labor market is affected by and linked to technological progress, as technological developments can have a direct impact on unemployment levels;
- **Labor supply is influenced by workers' behaviors and preferences:** Such as preferred working hours, desire for leisure time, and the level of humanity or worker treatment within the workplace.

2.3 Labor Market Indicators: There are numerous indicators that reflect the state of the labor market and measure its performance. Among these indicators, the following can be distinguished (djalal & ahmed, 2020, pp. 46-47):

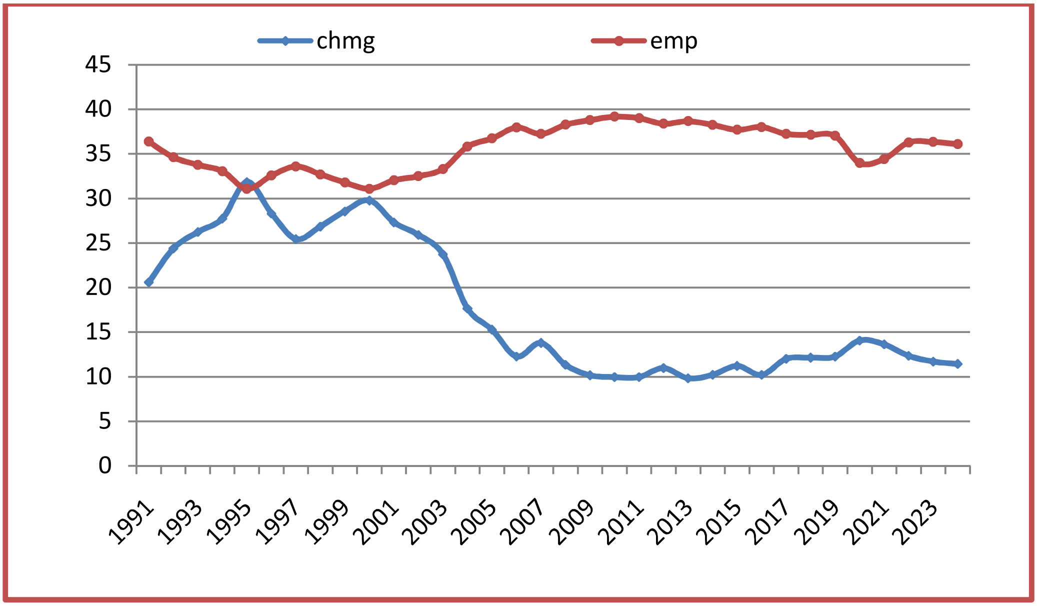
- Labor force participation rate;
- Employment rate;
- Employment status;
- Employment by sector;
- Employment by occupation;
- Part-time workers;
- Working hours;

- Employment in the informal sector;
- Overall unemployment;
- Time-related underemployment;
- Poverty, income distribution, and job opportunities by income categories and the working poor.

3. Analysis of the Labor Market Indicators and the Official Exchange Rate in Algeria (1991–2024)

The depreciation of the dinar is considered one of the key economic factors contributing to rising inflation rates, primarily due to increased import costs and higher prices for goods and services, particularly those linked to international markets. This rise in prices, in turn, reduces individuals' purchasing power, which weakens domestic demand and adds further pressure on the labor market, especially in sectors dependent on local consumption. As these pressures persist, employment opportunities decline, leading to higher unemployment rates and deepening socio-economic challenges. This dynamic will be examined through the analysis of the two figures presented below, which illustrate the evolution of both the dinar's value and unemployment rates over the period from 1991 to 2024, in order to better understand the interrelationship between these key economic variables during this time frame.

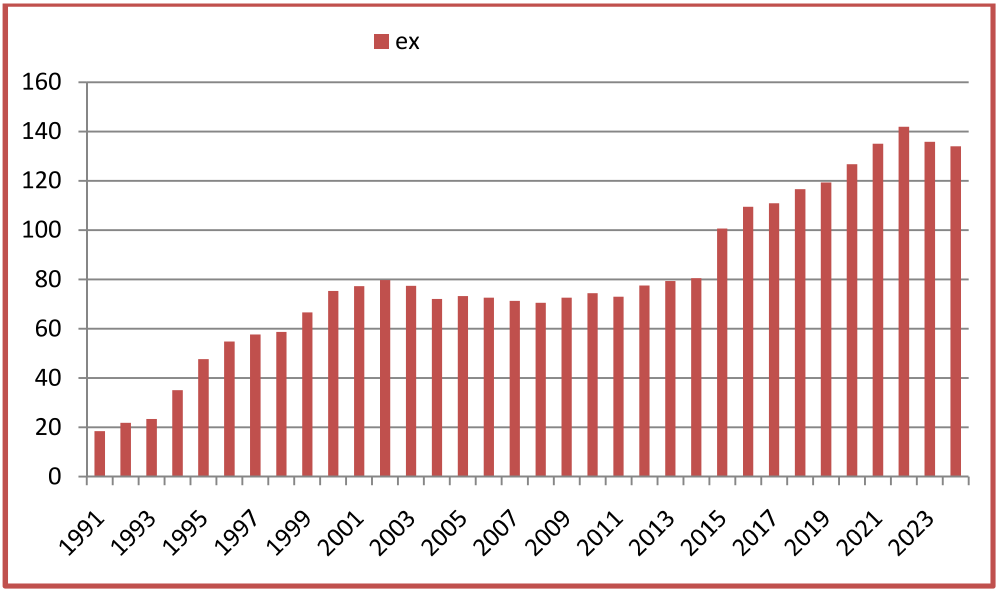
Fig.1: Analysis of the Labor Market Indicators in Algeria 1991–2024



Source: World Bank data.

The graph above illustrates notable trends in Algeria's labor market over the years. During the 1990s, the unemployment rate surged, reaching a peak of 31.84% in 1995—an indication of the severe economic and social challenges facing the country at the time. From the early 2000s onward, particularly after 2001, the unemployment rate began to decline steadily, hitting its lowest level of 9.82% in 2013. However, since 2014, it has shown fluctuations between 10% and 12%, with a significant spike to 14.06% in 2020 due to the impact of the COVID-19 pandemic on the labor market.In contrast, the employment rate remained relatively stable throughout the period, generally fluctuating between 31% and 39%. This suggests that job creation has not kept pace with the growth of the working-age population. Nevertheless, the period from 2021 to 2024 saw modest improvements in labor market conditions, with unemployment falling to 11.42% and employment rates showing a gradual upward trend.

Fig.2: Analysis of the Official Exchange Rate in Algeria 1991–2024



Source: World Bank data.

It can be observed from the above figure that the exchange rate of the Algerian local currency against the US dollar from 1991 to 2024 has shown a clear and continuous upward trend, reflecting a depreciation in the value of the Algerian dinar against the dollar. In the early 1990s, the exchange rate was relatively low, reaching about 18.47 dinars per dollar in 1991. It then began to rise sharply over the following years, reaching approximately 35 dinars per dollar in 1994, and continued to increase to around 77 dinars in 2001. From 2002 to 2014, the exchange rate experienced minor fluctuations with some decreases and increases, but the overall trend remained relatively stable around an average of 70-80 dinars per dollar. However, starting from 2015, the rate of increase accelerated significantly, with the exchange rate jumping from about 80 dinars to over 100 dinars per dollar. It continued to rise, reaching its highest recent level of approximately 141.99 dinars in 2022. This rising trend indicates a continuous decline in the strength of the local currency over the past three decades, with a notable increase in recent years that may be linked to local and global economic factors such as oil price fluctuations, inflation, and monetary policies. Despite some slight decreases after 2022, the exchange rate remains at high levels compared to the beginning of the period.

4. Measuring the Impact of Labor Market Indicators on the Exchange Rate in Algeria During the Period 1991–2024

After identifying and selecting several indicators that represent the labor market and influence the official exchange rate in Algeria 1991-2021, a multiple linear regression model was applied using the Eviews 10. The (OLS) method was used. The table below presents the variables used in the econometric model, which were collected from the World Bank database, as follows:

Table 1. Variables Used in the Econometric Model

Variable Name	Variable Code	Type of Variable
Official exchange rate (local currency per US dollar, period average)	EX	Dependent variable
Employment to population ratio, total (%)	EMP	Independent variable
Unemployment (% of total labor force)	CHMG	Independent variable
Youth unemployment (% of total labor)	CHMGG	Independent variable
Female youth unemployment (% of total labor)	CHMGF	Independent variable
Natural logarithm of gross capital formation (% of GDP)	GCC	Independent variable

Source: World Bank data.

3.1 Formulation of the Econometric Model:

The stage of building and formulating the econometric model is considered one of the most important steps in economic analysis. To estimate the model, the (OLS) method was used, as it provides linear and unbiased estimators. Therefore, OLS is regarded as one of the best methods for estimating linear models due to its desirable properties and assumptions when estimating a multiple linear regression model (bourahla, 2021, p. 123):

$$H_1 : E(\varepsilon_i) = 0 \qquad ,\forall i$$
$$H_2 : var(\varepsilon_i) = E(\varepsilon_i^2) = \sigma_u^2$$
$$H_3 : cov(\varepsilon_i \varepsilon_j) = E(\varepsilon_i \varepsilon_j) = 0 \qquad ,\forall\, i \neq j$$
$$H_4 : cov(\varepsilon_i X_i) = E(\varepsilon_i X_i) = 0$$
$$H_5 : \varepsilon_i \rightarrow N(0\,,\ \sigma^2\,)$$

After identifying the variables included in the econometric model and collecting the relevant data for each, the mathematical form of the model was defined as a function, as follows:

In order to study this function, the multiple linear regression method was used to estimate the econometric model for this study. The mathematical form of the model is written as follows:

EX

Where:

- **t:**the time;
- $\boldsymbol{\beta_0, \beta_1 \beta_2\,, \beta_3\,, \beta_4\,, \beta_5}$: the parameters of the model;
- $\boldsymbol{\varepsilon_t}$: the random error term or.

3.2 Model Estimation: After entering the data for the study variables represented by selected labor market indicators and the official exchange rate into the statistical software Eviews 10, the multiple linear regression model was estimated. The results are shown in the table below:

Table 02: Results of the Multiple Linear Regression Model Estimation

Dependent Variable: EX
Method: Least Squares
Date: 08/05/25 Time: 18:04
Sample: 1991 2024
Included observations: 34

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	400.0590	171.9402	2.326734	0.0274
EMP	-8.217327	3.914167	-2.099381	0.0449
CHMG	-9.139701	1.729243	-5.285376	0.0000
CHMGG	3.177547	1.578788	2.012649	0.0539
CHMGF	1.638677	0.602402	2.720241	0.0111
GCC	-1.078214	0.442119	-2.438742	0.0213

R-squared	0.936708	Mean dependent var	80.64108
Adjusted R-squared	0.925406	S.D. dependent var	33.00539
S.E. of regression	9.014386	Akaike info criterion	7.394306
Sum squared resid	2275.256	Schwarz criterion	7.663664
Log likelihood	-119.7032	Hannan-Quinn criter.	7.486164
F-statistic	82.87924	Durbin-Watson stat	1.306189
Prob(F-statistic)	0.000000		

Source: Prepared by the researcherbased on Eviews 10.

Accordingly, based on Table 02, the model is written according to the following equation:

$$\widehat{EX} = 400.0590 - 8.217327 * \mathrm{EMP} - 9.139701 * \mathrm{CHMG} + 3.177547 * \mathrm{CHMGG} + 1.638677 * \mathrm{CHMGF} - 1.078214 * \mathrm{GCC}$$

$$Pro_{\hat{\beta}_i}: (0.0274)(0.0449)(0.0000)(0.0539)(0.0111)(0.0213)$$

$$R^2 = 0.936708 \qquad \overline{R^2} = 0.925406$$

$$F_c = 82.87924 \qquad Pro_F = 0.000000 \qquad n = 34$$

3.2.1 Statistical Analysis: From the results of the model estimation, we observe the following:

- The variable Total Employment Rate (as a percentage of the population) demonstrates statistical significance, as its p-value is

0.0449 below the conventional threshold of 0.05. This result leads us to reject the null hypothesis (H_0) in favor of the alternative (H_1), indicating that the employment rate exerts a meaningful influence on the official exchange rate.

- The coefficient of the variable Unemployment Rate (% of total labor force) is also statistically significant since its p-value is less than 0.05, specifically$Pro_{\widehat{\beta}_1} = 0.0000$. Hence, we reject $H_0$and accept H_1, indicating it affects the official exchange rate.
- The coefficient of the variable Youth Unemployment (% of total labor force) is statistically significant, with a p-value slightly above 0.05$Pro_{\widehat{\beta}_1} = 0.0539$. Thus, we reject $H_0$and accept H_1, meaning it influences the official exchange rate.
- The p-value associated with the coefficient of Female Youth Unemployment (% of total female labor force) is 0.0111, indicating statistical significance at the 5% level. Hence, we reject the null hypothesis and accept the alternative, confirming its influence on the official exchange rate.
- The coefficient of the variable Gross Capital Formation (% of GDP) is not statistically significant, as its p-value is greater than 0.05, $Pro_{\widehat{\beta}_1} = 0.0213$. Thus, we accept $H_0$and reject H_1, meaning it does not have a significant effect on the official exchange rate.
- The significance of the entire model is confirmed by the F-test, yielding a p-value of 0.000000, far below the 0.05 threshold. Consequently, the null hypothesis is rejected, affirming that the independent variables jointly exert a meaningful impact on the dependent variable

3.2.2 Economic Analysis: Based on the estimation results, the model can be economically analyzed as follows:

- The value of (R^2) that the proposed multiple linear regression model represents the relationship under study well, as 93.67% of the variations in the official exchange rate are explained by employment, while the remaining 6.33% is attributed to unspecified factors represented by the random error term.
- The coefficient of the Total Employment Rate (as a percentage of the population) is negative, indicating an inverse relationship between total

employment and the official exchange rate. Specifically, a 1% increase in the total employment rate leads to a 8.217327% decrease in the official exchange rate.

- An inverse association exists between the unemployment rate and the official exchange rate, as evidenced by the negative coefficient. Each 1% uptick in unemployment results in an approximate 9.14% fall in the official exchange rate.
- The coefficient of the variable Youth Unemployment (% of total male labor force) is positive, indicating a direct relationship with the official exchange rate. A 1% increase in male youth unemployment results in the official exchange rate increasing by 3.177547%.
- The coefficient of the variable Youth Unemployment (% of total female labor force) is also positive, showing a direct relationship. A 1% increase in female youth unemployment leads to the official exchange rate increasing by 1.638677%.
- The coefficient of Gross Capital Formation (% of GDP) is negative, reflecting an inverse relationship with the official exchange rate. A 1% increase in gross capital formation corresponds to a 1.078214% decrease in the official exchange rate.

3.3 Standard Tests for the Multiple Linear Regression Equation:

After confirming the model's statistical and economic validity, we proceed to test it from a diagnostic standpoint to assess its compliance with underlying assumptions. Many economists believe that the Durbin-Watson statistic does not provide accurate results, especially in small samples, as it may fail to detect autocorrelation in the error terms, particularly in ambiguous cases. Therefore, we rely on several tests including the LM test, ARCH test, Jarque-Bera test, in addition to assessing the model's predictive power as follows:

Table 3. LM Test

Breusch-Godfrey Serial Correlation LM Test:

F-statistic	2.907252	Prob. F(2,26)	0.0725
Obs*R-squared	6.213931	Prob. Chi-Square(2)	0.0447

Source: Prepared by the researcherbased on Eviews 10.

According to the results in Table 03, the computed F-statistic of 2.907252 with a p-value of 0.0725 exceeds the 5% critical value, indicating that residuals are free from serial autocorrelation and the model specification is statistically robust in this regard

Table 4. ARCH Test

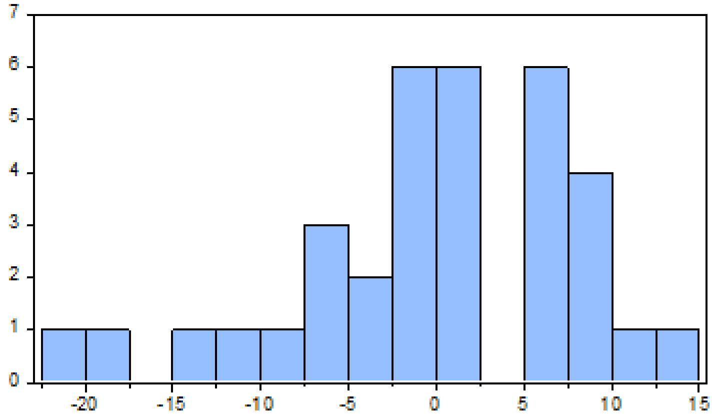
Heteroskedasticity Test: ARCH

F-statistic	0.484593	Prob. F(1,18)	0.4952
Obs*R-squared	0.524321	Prob. Chi-Square(1)	0.4690

Source: Prepared by the researcherbased on Eviews 10

As indicated in Table 04, the ARCH test for heteroskedasticity yields an F-statistic of 0.215451 and a p-value of 0.6458. Given that the p-value exceeds the 0.05 significance level, the null hypothesis of homoskedasticity is upheld. Thus, the residuals display no evidence of conditional variance instability

Fig.3: Normality Test of the Model Residuals



Series: Residuals	
Sample 1991 2024	
Observations 34	
Mean	1.87e-13
Median	0.205080
Maximum	14.24065
Minimum	-21.99203
Std. Dev.	8.303443
Skewness	-0.712503
Kurtosis	3.374472
Jarque-Bera	3.075400
Probability	0.214875

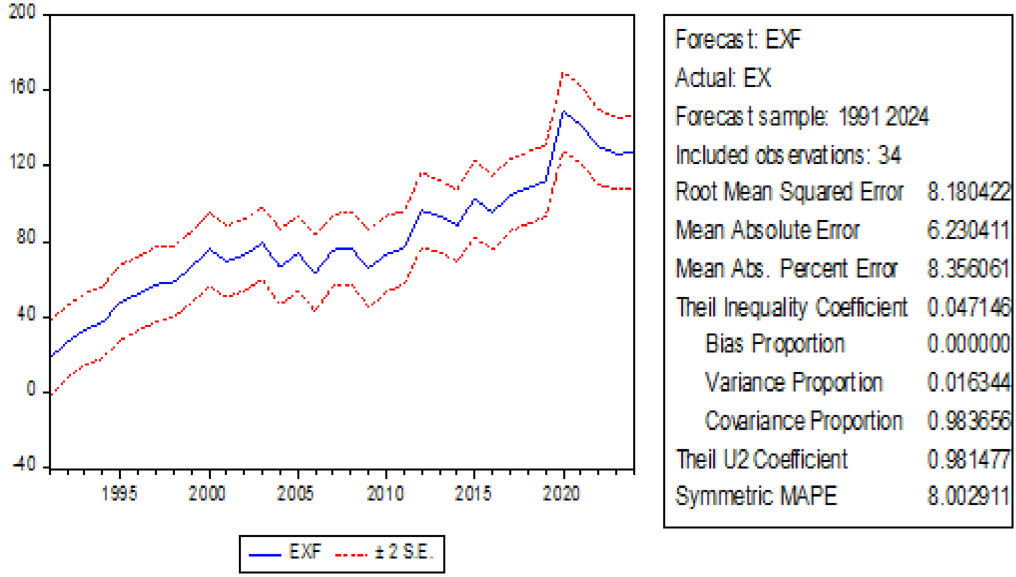
Source: Prepared by the researcherbased on Eviews 10

Figure 03 presents where the p-value is 0.214875exceeding the 5% significance threshold. This supports the acceptance of the null hypothesis, indicating that the residuals are normally distributed.

3.4 Testing the Predictive Power of the Model:

The predictive power of the model can be tested using Theil's Inequality Coefficient, as illustrated in the figure below.

Fig.4: Illustration of Theil's Inequality Coefficient Test



Source: Prepared by the researcherbased on Eviews 10.

From Figure No. 04, we observe that the estimated model has an acceptable predictive power, as indicated by Theil's Inequality Coefficient, which equals 0.047146. Since this value is close to zero, it suggests that the model as a whole has an acceptable predictive ability for the official exchange rate.

4. CONCLUSION

This study aimed to conduct a theoretical approach to examine the relationship between labor market indicators and the official exchange rate in Algeria. It addressed the key concepts related to labor market indicators, their characteristics and features, as well as the most important indicators on which the analysis was based. In addition, it highlighted the impact of labor market indicators on the official exchange rate in Algeria during the period from 1991 to 2024, through an empirical study using the multiple linear regression model.

4.1 Study Findings: Among the most important findings of the study, the following points can be highlighted:

- Algeria is still considered one of the countries suffering from unemployment, as the demand for jobs by individuals exceeds the supply in the labor market.

- In Algeria, employment is negatively associated with the official exchange rate. Specifically, a 1% rise in employment results in an 8.217327% decline in the official exchange rate.
- Consistent with economic expectations, the unemployment rate negatively influences the official exchange rate a 1% increase in unemployment leads to an approximate 9.14% depreciation, reflecting the broader economic slowdown associated with labor market weakness.
- Male unemployment, on the other hand, shows a positive correlation with the official exchange rate; a 1% rise in male unemployment corresponds to a 3.177547% increase in the exchange rate.
- A comparable positive relationship exists between female unemployment and the official exchange rate, where a 1% increase in female unemployment leads to a 1.638677% rise in the exchange rate.
- Gross capital formation negatively affects the official exchange rate, with a 1% increase in capital formation contributing to a 1.078214% decrease in the exchange rate.

4.2 Recommendations: The following recommendations were derived from this study:

- The Algerian government should reduce unemployment by adopting macroeconomic policies aimed at lowering poverty rates in Algeria.
- There is a need to create job opportunities in productive economic sectors rather than relying solely on the administrative and service sectors.
- Arab economies, with Algeria at the forefront, need to embed labor market balancing mechanisms into their development agendas by addressing both demand- and supply-side challenges .
- It is important to foster competition among private companies and encourage their geographic expansion by increasing their branches throughout Algeria. This can be done by offering opportunities to foreign investors to establish their own projects, thus contributing to higher economic growth rates.

- There is a need for economic diversification and for moving away from dependence on the hydrocarbons sector.
- Algeria must strive to benefit from leading international experiences in the field of employment.

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